

Appendix E. Sensitivity analysis and robustness of trap rupture conditions

This appendix verifies mechanism robustness by systematically varying parameters around the Scenario I baseline. We organize results into dual core levers reconfiguring accountability and verification, and auxiliary levers amplifying downstream feedback.

Appendix E.1. Dual core levers: Governmental penalty (f_p) and in-stream audit efficiency (u_2)

Analysis confirms governmental penalties and platform technology as the primary levers for systemic change.

- **Governmental Penalty (f_p):** Starting from $E_2(0, 0, 1)$ ($f_p = 30$), increasing f_p reveals a threshold at $f_p = C_{p2} = 80$. Above this threshold, passive governance becomes prohibitive. The variable f_p is the sole institutional lever directly internalizing accountability, rendering passivity a dominated strategy.
- **In-Stream Audit Efficiency (u_2):** Increasing u_2 from 0.4 reveals a threshold at $u_2 \approx 0.65$, where illicit returns become negative. Verification improvements yield limited benefits below the threshold; however, crossing the critical value makes compliance self-enforcing.

Appendix E.2. Auxiliary user incentives (C_u and f_u)

User incentives affect downstream feedback volume rather than upstream accountability provision, consistent with our interpretation.

- **Reducing User Reporting Cost (C_u):** Reducing C_u (even to 0) increases participation but fails to reconfigure the trap's defining accountability allocation; the system reverts to E_2 .
- **Increasing User Compensation (f_u):** Higher rewards stabilize γ but do not compel platform proactivity. This confirms that f_u primarily affects feedback signals, while strategic shifts depend on structural levers f_p and u_2 .

Appendix E.3. Robustness check on initial conditions

To address sensitivity of the evolutionary dynamics to initial conditions, we conducted robustness checks using multiple initial condition sets representing different early-stage ecosystem states.

We tested the following scenarios alongside our baseline:

- $(\alpha_0, \beta_0, \gamma_0) = (0.3, 0.5, 0.4)$ (Baseline: early-stage ecosystem)
- $(\alpha_0, \beta_0, \gamma_0) = (0.1, 0.3, 0.2)$ (Low engagement scenario)
- $(\alpha_0, \beta_0, \gamma_0) = (0.5, 0.5, 0.5)$ (Medium engagement scenario)
- $(\alpha_0, \beta_0, \gamma_0) = (0.7, 0.7, 0.6)$ (High engagement scenario)

As illustrated in Figure E.10, the qualitative conclusions remain invariant across all perturbations. Regardless of the starting point, the system converges to the same evolutionary stable strategy (ESS) under the given parameters (e.g., system exit requires $f_p > C_{p2}$ and/or sufficient u_2). This dynamic invariance validates Equation 8 as a generalized mechanism summary, confirming that the user-led content governance trap is a structural attractor rather than an artifact of specific initial values.

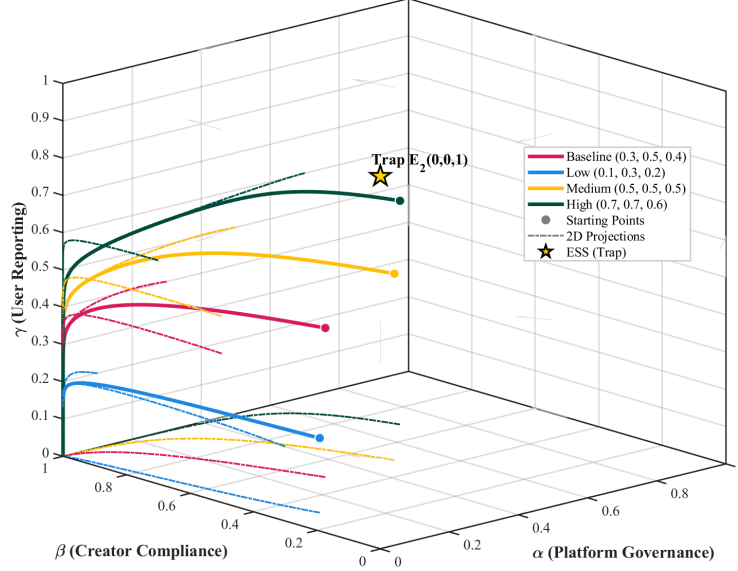


Figure E.10: Evolutionary trajectories under different initial conditions, demonstrating robust convergence to the same equilibrium state.